



One Pager Product Proposal: Contactless Radar-Based Heart Rate Monitoring Device

Electronics Project

Introduction & Problem Statement

Monitoring heart rate is essential in healthcare, fitness, and safety applications. Most commonly used solutions rely on wearable devices (e.g., smartwatches or chest straps) that require direct contact with the body. However, this approach is not always practical or feasible, for example, in long-term monitoring, sleep tracking, or when dealing with sensitive or immobile patients.

A contactless solution could significantly improve user comfort, simplify measurements, and enable new applications in environments where wearable devices are impractical.

System Concept

Based on established research on Doppler radar physiological sensing [1], this project proposes a **contactless heart rate monitoring system** using microwave Doppler radar. A conceptual illustration of the system setup is shown in *Fig. 1* (see Appendix A). The system is based on the **K-LC5 Doppler radar module** and an **STM32F4 microcontroller**. The radar detects micro-movements of the human chest caused by cardiac activity through Doppler-induced phase modulation of the reflected signal. The STM32F4 processes the radar output signal by applying filtering and frequency-domain analysis, and extracts the heart rate in real time.

Key features of the system include:

- **Contactless operation:** No physical sensors attached to the body
- **High sensitivity:** Detection of sub-millimeter chest movements
- **Measurement range:** Effective operation at short distances (typically up to 1 m)
- **Real-time processing:** On-board signal processing using the STM32F4 platform
- **Compact embedded system:** Suitable for integration into various environments

Although the measurement accuracy is expected to be lower than that of medical-grade ECG systems, the proposed solution provides sufficient precision for **continuous monitoring and trend analysis** [2].

Development Plan

The implementation of the system requires:

- Design of an **analog front-end** for radar signal conditioning
- **Integration of the K-LC5 radar module** with the STM32F4 Discovery board
- Signal acquisition using the STM32 ADC
- **Firmware development** for filtering, respiration suppression, and heart rate extraction
- **Visualization of measurement data** via a suitable interface
- Experimental **validation** through comparison with a reference ECG heart rate sensor
- A **dielectric lens** may be used to focus the radar beam and improve signal quality

Appendix A – System Illustration

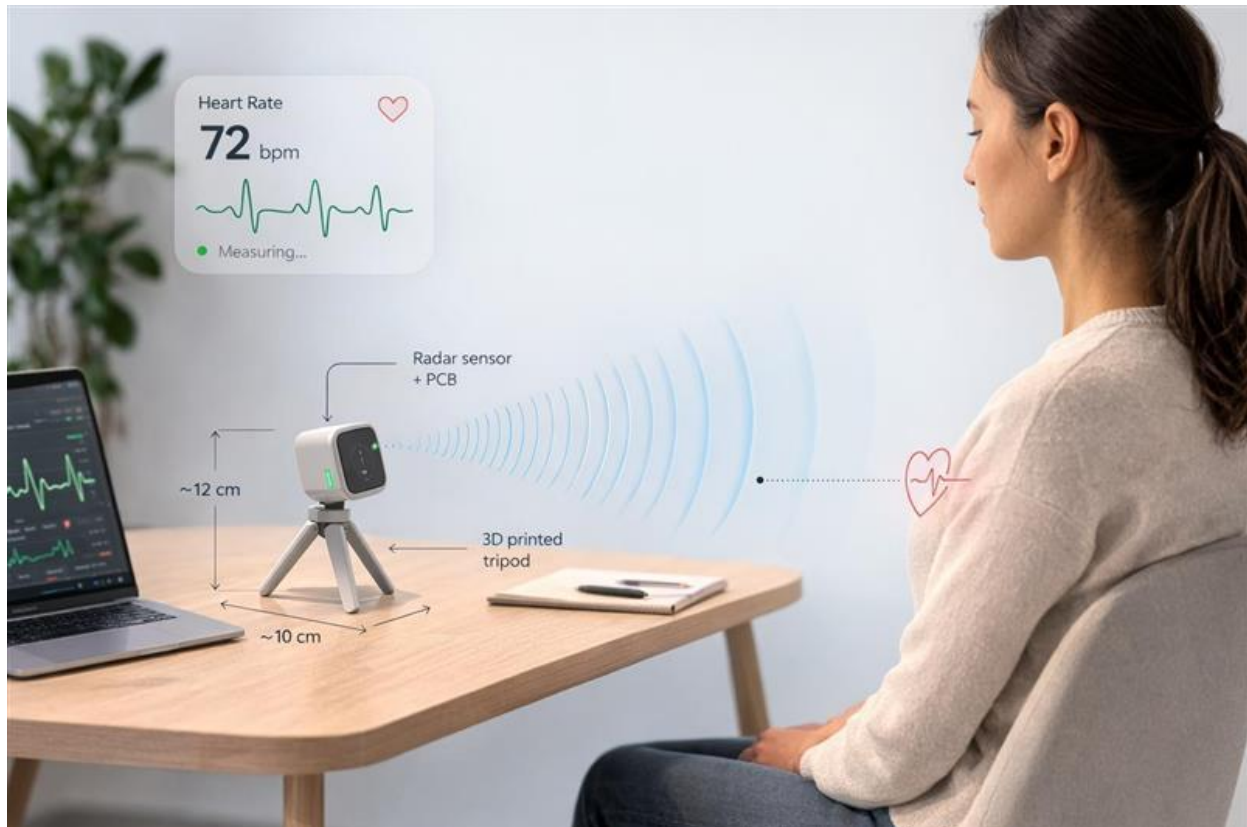


Figure 1. Conceptual illustration of the proposed radar-based heart rate monitoring system setup.
The image was generated with the support of a generative AI tool and adapted by the authors.

Declaration of AI

It should be noted that generative AI systems were used as supporting tools during the preparation of this project proposal. In particular, ChatGPT was used for language refinement, clarification of technical concepts, assistance with structuring the document, and for generating illustrative schematic images of the proposed system setup.

References

- [1] O. Boric-Lubecke, V. M. Lubecke, A. D. Droitcour, B.-K. Park, and A. Singh, *Doppler radar physiological sensing*, 1st ed. in Wiley series in biomedical engineering and multi-disciplinary integrated systems. Hoboken, New Jersey: IEEE, Wiley, 2016. doi: 10.1002/9781119078418.
- [2] S. Yuan, S. Fan, Z. Deng, and P. Pan, "Heart Rate Variability Monitoring Based on Doppler Radar Using Deep Learning," *SENSORS-BASEL*, vol. 24, no. 7, pp. 2026-, 2024, doi: 10.3390/s24072026.